NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 30/03/2026
Time: 3 Hrs | GTM – 09 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	C	B	A	B	B	C	D	B	A
11	12	13	14	15	16	17	18	19	20
A	B	D	D	C	A	A	C	B	A
21	22	23	24	25					
3	195	40	50	5					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	C	D	C	B	C	B	A	D	B
36	37	38	39	40	41	42	43	44	45
D	B	D	C	D	C	C	C	C	B
46	47	48	49	50					
4	9	15	40	2					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	B	B	B	A	B	C	B	B	B
61	62	63	64	65	66	67	68	69	70
A	C	C	C	B	D	A	B	A	A
71	72	73	74	75					
2	17	1	196	28					

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HINTS
PHYSICS

1. Effective force constant in case (iii) and (iv) is
$K_{eff} = 2\ k + 2\ k = 4\ k$

Therefore, $T_1 = T_2 = 2\pi\sqrt{\frac{m}{k}}$ and $T_3 = T_4 = 2\pi\sqrt{\frac{m}{4k}} = \pi\sqrt{\frac{m}{k}}$

2. $(l_1 + l_2)\alpha_{eq}\Delta\theta = l_1\alpha_1\Delta\theta + l_2\alpha_2\Delta\theta$

$\alpha_{eq}\ \frac{l_1\alpha_1 + l_2\alpha_2}{l_1 + l_2}$

3. Apparent weight W_a = Actual weight (w) – upthrust (F)

Here, $F = V_{PW}g$ (ρ w = density of water)

i.e. $F_{0^oC=} V_{0^oC=}(\rho_w)_{0^oC}g$

and $F_{50^oC=} V_{50^oC=}(\rho_w)_{50^oC}g$

$\therefore \frac{F_{50^oC=} V_{50^oC=}(\rho_w)_{50^oC}}{F_{0^oC=} V_{0^oC=}(\rho_w)_{0^oC}} = \frac{(1+\gamma_m\Delta\theta)}{(1+\gamma_w\Delta\theta)}$

Given that $\gamma_m < \gamma_w$ $\therefore F_{50^oC} < F_{0^oC}$

Or apparent weight at 50^oC will be more

4. The time constant of the CR circuit is $\tau = CR = 10^{-3}$ s. then we have

$q(t) = CV[1-\exp(-t/\tau)]$

$= 2x10^{-9}[1-\exp(-t/10^{-3})]$

The electric field in between the plates at time t is

$E = \frac{q(t)}{\varepsilon_0 A}$, $A = \pi(0.5)^2\ m^2$ = area of the plates.

Consider now a circular loop of radius (1/4)m parallel to the plates passing through P. The magnetic field B at all points on the loop is along the loop and of the same value. The flux through this loop is given by

$\Phi = E\ \pi\left(\frac{1}{4}\right)^2 = \frac{\pi E}{18}$

The displacement current

$i_d = \varepsilon_0\frac{d\Phi}{dt} = \frac{1}{4}\frac{dq}{dt} = 0.5x10^{-6}\exp(-1)$ at t= 10^{-3} s. Now,

Applying Ampere-Maxwell law to the loop, we get

$B \times 2\pi \times \left(\frac{1}{4}\right) = \mu_0(i_c + i_d) = \mu_0(0 + i_d)$

$= 0.5x10^{-16}\ \mu_0\exp(-1)$

Or $B = 1.48x10^{-13}T$

5. $eV = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$

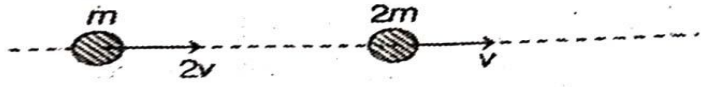
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e $\frac{V}{3}=\frac{hc}{2\lambda}-\frac{hc}{\lambda_0}$ Solving these two equations we get $\lambda_0=4\lambda$

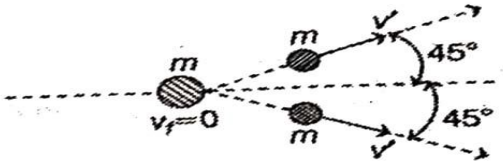
6. Energy released = Final binding energy
- initial binding energy
= 110x8.2+90x8.2-200x7.4
= 160 MeV

7. C
If $V_A > V_B$, both the junction diodes are forward biased and the given circuit diagram becomes a balanced Wheatstone bridge. The equivalent resistance in this case becomes 4 Ω if $V_A < V_B$, the diodes are reverse biased in that case 4 Ω , 5 Ω and 4 Ω are in series.

8. According to the question, Initial condition,



Final condition



AS we know that, in collision linear momentum is conserved in both x and y directions separately
So,

$(P_x)_{initial}=(P_x)_{final}$

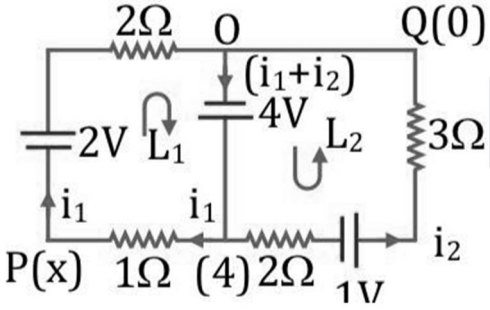
$m(2v)+2m(v)=0+mv'\cos 45^0+mv'\cos 45^0$

$\Rightarrow 4mv=\frac{2m}{\sqrt{2}}v'\Rightarrow v'=2\sqrt{2}\,v$

So, each particle will move with a speed of $2\sqrt{2}\,v$

9. Loop 1 $\rightarrow 4-i_1+2-2i_1=0$

$i_1=2A$



$V_P+2-2(2)=V_Q$

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$V_P - V_Q = 2A$

10. EMF induced in the triangular loop is given as loop is given as $e = \frac{d\phi}{dt}$

$\Rightarrow e = A\frac{dB}{dt} \Rightarrow e = \sqrt{3} \times \sqrt{3} = 3V$

By symmetry in each rod emf will be IV. The current in the triangular loop is given as

$i = \frac{e}{R} = \frac{3}{5} = 0.6A$

Voltage drop between A and B $= 0.6 \times 1 = 0.6V$

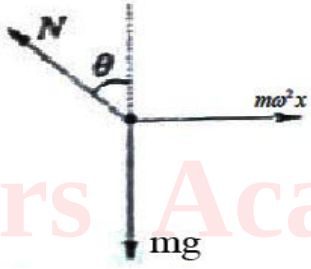
So voltage across AB is given as

$V_{AB} = 1 - 0.6 = 0.4V$

11. Use principle of method of mixture

12. $\tan\theta = \frac{\omega^2 x}{g}$

$\frac{dy}{dx} = \frac{\omega^2 x}{g} \Rightarrow y = \frac{\omega^2 x^2}{g}$

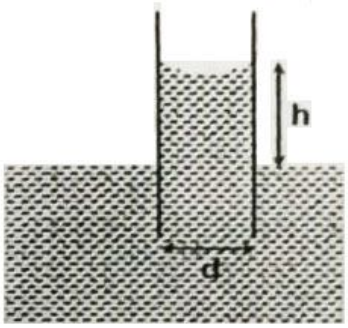


13. $2T\ell = \rho d\ell hg$

$h = \frac{2T}{\rho g d} = \frac{2 \times 7 \times 10^{-2}}{10^3 \times 10 \times 1 \times 10^{-4}} = 14cm$

Force of mutual attraction between the plates is

$F = \int_0^h \rho g y \ell dy = \frac{\rho g \ell h^2}{2} = \frac{10^3 \times 10 \times 0.10 \times \left(14 \times 10^{-2}\right)^2}{2} = 9.8N$



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14. $m \propto t^a v^n \ell^c$

$m \propto [T]^a [LT^{-1}]^b [ML^2T^{-1}]^c$

$M^1L^0T^0 = M^c L^{b+2c} T^{a-b-c}$

Comparing powers $c = 1, b = -2, a = -1$

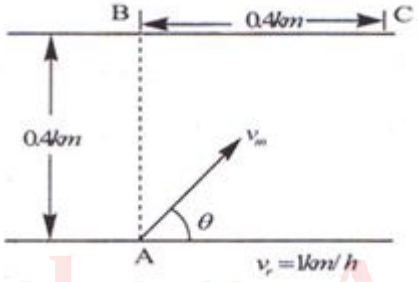
$m \propto t^{-1} v^{-2} \ell^1$

15. $v_m \sin\theta = \frac{0.4}{t}$ .....(1)

$v_m \cos\theta + v_r = \frac{0.4}{t}$ ..... (2)

From (1) *and* (2)

$v_m \sin\theta = v_m \cos\theta + v_r$



$5\sin\theta = 5\cos\theta + 1$

$5\sqrt{1-\cos^2\theta} = 5\cos\theta + 1$

$25(1-\cos^2\theta) = 25\cos^2\theta + 1 + 10\cos\theta$

$50\cos^2\theta + 10\cos\theta - 24$

$\cos\theta = \frac{3}{5}, -\frac{4}{5}$

$\theta = 53^\circ$

16. $B_0 = \frac{E_0}{C} = \frac{300}{3x10^8} = 10^{-6} T$

$F_{max} = qvB_0 = 1.6x\ 10^{-19} x\ 2\ x10^7\ x\ 10^{-6} = 3.2x10^{-18}\ N$

17. $\frac{1}{2} KA^2 = 1.0J$

$\therefore \quad K = \frac{2}{A^2} = \frac{2}{(0.4)^2} \frac{25}{2} N/m$

$T = 2\pi\sqrt{\frac{m}{K}} = 2\pi\sqrt{\frac{2}{(25/2)}} = \frac{4\pi}{5} S$

= 2.51 s

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18. Resolve the dipole along OP and perpendicular to OP. Find the fields due to these two and add vectorially.

19. The equation of fusion is represented as ${}_1H^2 + {}_1H^2 \rightarrow {}_2He^4 + Energy$
The binding energy per nucleon of deuteron is 1.1 MeV. Therefore, the net initial binding energy $= 4 \times 1.1 = 4.4MeV$. The binding energy per nucleon of ${}_2He^4$ *is* 7.0 *MeV*. Therefore the net final binding energy is $= 4 \times 7.0 = 28.0MeV$. Hence, energy released $= 28.0MeV - 4.4MeV = 23.6MeV$

20.

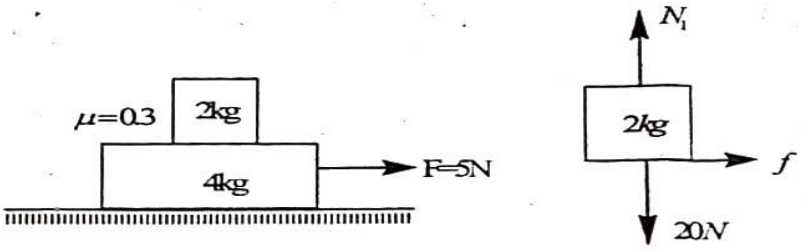
21. As the expression is a third order expression in degree of sinusoidal function hence it is a superposition of three simple harmonic motions

22. The charge on $12\mu F$ is
$q = (12\mu)(60V) = 720\mu C$
This will be common on all of them.
So net potential difference
$$= \frac{q}{C_{eff}} = \frac{720\mu C}{\left(\frac{48}{13}\mu\right)} \Rightarrow \frac{(720\mu C)}{48\mu} \times 13 \Rightarrow 195V$$

23. $$\frac{dA}{dt} = \frac{L}{2m} = \frac{mvr}{2m} = \frac{vr}{2} \quad \therefore \quad v = \frac{2(dA/dt)}{r} = \frac{2x40x10^{16}}{2x10^{12}}$$
$= 4 \times 10^4$ m/s**= 40 km/s**

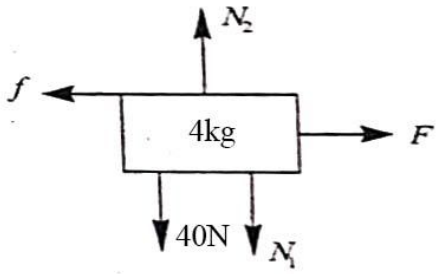
24. For coil R = 200/1 = 200 Ω
With a.c. $i = \frac{200}{\sqrt{R^2 + x_L^2}}$

25.



Limiting friction between blocks is $f_{max} = \mu N_1 = 0.3 \times 20 = 6N$
To start sliding between blocks we use
$$f_{max} = 2\left(\frac{F}{6}\right) = 6 \Rightarrow F = 18N$$
$\Rightarrow$ at F = 15N both blocks move together at $a = \frac{F}{6} = \frac{5}{2} m/s^2$
$$\Rightarrow f = 2a = 2 \times \frac{5}{2} = 5N$$

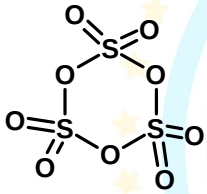
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CHEMISTRY

26. CO is strong field ligand so pairing of electrons occurs, in case of Pt and Pd paring occurs.

27. IUPAC name of compound is Diammine silver (I) dicyanidoargentate (I)

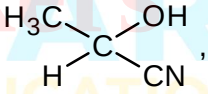
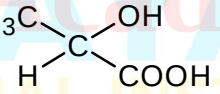
28. 

29. The formula of brown ring is $\left[Fe(NO)(H_2O)_5\right]SO_4$

30. $Cu^{2+} + H_2S \longrightarrow CuS \xrightarrow{HNO_3} S + NO + Cu(NO_3)_2$

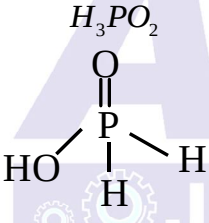
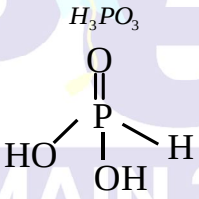
$Cu(NO_3)_2 \xrightarrow{NH_3} \left[Cu(NH_3)_4\right]SO_4$

31. hydrazine does not contain Carbon & azobenzene lose N_2.

32. CH_3CHO,  ,  only matches with optical activity and haloform conditions.

33.

$\Delta_t H_1$	801	577	579	558	589

34.  

35. $\Delta E = -2.18\times10^{-18}\left[\frac{1}{2^2}-\frac{1}{5^2}\right]$

$= -2.18\times10^{-18}\left[\frac{25-4}{100}\right]$

$= -2.18\times10^{-18}\left[\frac{21}{100}\right]$

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$= 2.18\times10^{-20}\times21 = -4.578\times10^{-19}$ J

36. As_2O_3 is amphoteric in nature

37. Conceptual

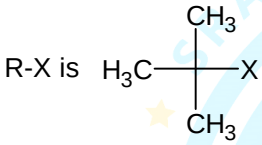
38. Based on MOED

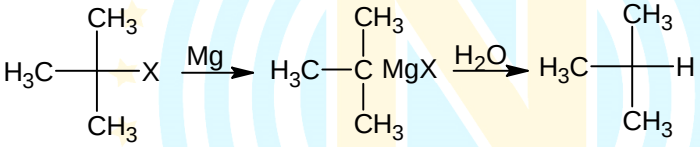
39. $\Delta H = C_P\left[H_2O(l)\right]\Delta T + \Delta H_{freezing} + C_P\left[H_2O_{(S)}\right]\Delta T$

$= 75.3(0-10) + \left(-6.03\times10^3\right) + 36.8(-10-0)$

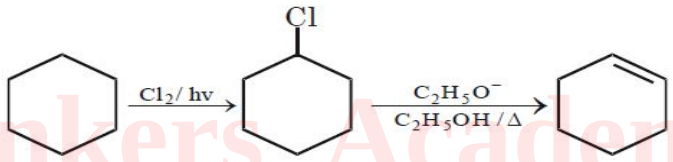
$= -753 - 6030 - 368$

$= -7151 J.mol^{-1} = -7.151 KJ\ mol^{-1}$

40. R-X is 



41. $Latose + H_2O \rightarrow \beta - D - Galactose + \beta - D - Glu\cos e$

42. 

43. Basic strength $\propto$ electron density

44. $CH_3COOH \xrightarrow{NaOH} CH_3COONa \xrightarrow{Electrolysis} C_2H_6 \xrightarrow[(h9)]{Cl_2(1mole)} C_2H_5Cl \xrightarrow{alcKOH} C_2H_4 \xrightarrow{Br_2} C_2H_4Br_2$

45. $K = \frac{2.303}{t}\log\frac{a}{a-x}$

$K = \frac{2.303}{(60-0)}\log\frac{1.24\times10^{-2}}{0.20\times10^{-2}}$

$= \frac{2.303}{60}\log\frac{124}{20}$

$= \frac{2.303}{60}(0.79) \qquad = 0.03032\,\text{min}^{-1}$

46. $2MnO_4^- + 3H_2O_2 \longrightarrow 2MnO_2 + 3O_2 + 2OH^- + 2H_2O$

3ml mole H_2O_2 react with 2 ml mole $KMnO_4$

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Then 6 ml mole H_2O_2 react with $\frac{2}{3}\times 6 = 4\,ml\,mole$

47. Basic buffer, Henderson equation

48. $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$

Moles $\frac{10}{17}$ $\frac{20}{32}$

Oxygen is a limiting reactant

$5\ mols\,O_2 \equiv 4\ moles\ NO$

$\frac{20}{32} moles\,O_2 \equiv \frac{4}{5}\times\frac{20}{32} moles\ NO$

Mass of NO = $\frac{4}{5}\times\frac{20}{32}\times 30 = 15\,gm$

49. Equivalent of OH^- produced $= \frac{2\times 20\times 60}{96500} = \frac{24}{965}$

Equivalent of NaOH produced

$= \frac{24}{965}$ = Equivalent of HNO_3 needed

$\frac{24}{965} = 0.62\times\frac{V}{1000}$

V=40Ml

50. $\%C = \frac{x\times 12\times 100}{54}$

$x = 4$

$\%H = \frac{Y\times 1\times 100}{54}$

$y = 6$

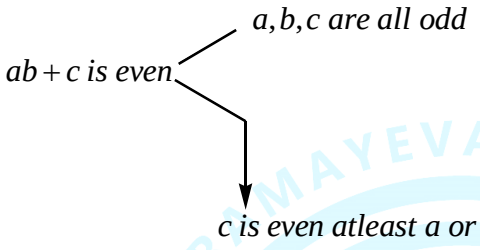
$DU = (4+1) - \frac{6}{2}$

$= 2$

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MATHEMATICS

51. P(number chosen is odd) = 3/5
P(number chosen is even)=2/5



E: $(ab+c)$ is even note that even E can occurs in two cases

E_1 :all the three numbers a,b and 'c' are odd $P(E_1)=\left(\frac{3}{5}\right)^3=\frac{27}{125}$

E_2 : 'c' is even and at least one of a or b is even $P(E_2)=\frac{2}{5}\left(1-\frac{9}{25}\right)=\frac{2}{5}.\frac{16}{25}=\frac{32}{125}$

$P(E)=P(E_1 \text{ or } E_2)$

$=P(E_1)+P(E_2)=\frac{59}{125}$

52. $f''(x)$ is an odd function

$\therefore \phi(\alpha,\beta)=0 \Rightarrow (\sin^4\alpha-1)^2+(\cos^4\beta-1)^2+2(\sin^2\alpha-\cos^2\beta)^2=0$

$\Rightarrow \sin^2\alpha+\sin^2\beta=1$

53. $f(x)=\frac{\sqrt{x^2-25}}{4-x^2}+\log_{10}(x^2+2x-15)$

For domain : $x^2-25\geq 0 \rightarrow x\in(-\infty,-5]\cup[5,\infty)$

$4-x^2\neq 0 \Rightarrow x\neq\{-2,2\}$

And $x^2+2x-15>0 \Rightarrow (x+5)(x-3)>0$

$\Rightarrow x\in(-\infty,-5)\cup(3,\infty)$

$\therefore x\in(-\infty,-5)\cup(5,\infty)$ is the domain

Hence $\alpha=-5;\ \beta=5 \Rightarrow \alpha^2+\beta^2=150$

54. $e=\sqrt{\frac{a^2+2-a^2-1}{a^2+2}}=\frac{1}{\sqrt{6}}$

55. $3\sin 3x+12\sin^3 x=a$
$\Rightarrow 9\sin x=a$

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$\Rightarrow a \in (-\infty, -9) \cup (9, \infty)$

56. $|a| = |b| = |c| = |d| = 1$

$|2a - 9d|^2 = |-6b - 7c|^2$

$\Rightarrow 3\cos\alpha(\alpha + \delta) = 7\cos(\beta + \gamma)$

57. $f(x) = 0, 1$ or x^k

58. Point of intersection $(-2, -2, -5)$ image of $(1, 2, -2)$ in the plane $2x + 5y - 4z - 6 = 0$ is $\left(\frac{-11}{45}, \frac{-10}{9}, \frac{22}{45}\right)$

$\Rightarrow$ equation of line is $\frac{x+2}{79} = \frac{y+2}{40} = \frac{z+5}{247}$

59. $(-Z)^r \hat{i} + z^{3s} \hat{j}$

$z^r \hat{i} + \hat{j} + z^r \hat{k}$

$\Rightarrow$ r is odd & 3 multiple and s can be any number

60. $e^x \geq x + 1 \forall x \in [0,1]$

$\Rightarrow \int_0^1 \frac{e^x dx}{x+1} > 1$ & both are increasing functions and $\frac{e^x}{x+1} - \frac{(x+1)}{2} > 0 =$ exact one solution

61. $a_n + a_{n+2} = 2a_{n+1} \Rightarrow f(x) = 0$

62. $K = \frac{4}{(2x-3)(3x-5y)} = \frac{7}{(y-2)(5y-7z)} = \frac{7}{(3z+1)(7z-3x)}$

$\frac{4}{(2x-3)} + \frac{7}{(y-2)} + \frac{7}{(3z+1)} = 0$

63. $2f(x).f'(x) = f(x).\frac{\cos x}{2 + \sin x}$

$2f'(x) = \frac{\cos x}{2 + \sin x}$

$2f(x) = \ln(2 + \sin x)$

$f(x) = \frac{1}{2}\ln\left(\frac{2 + \sin x}{2}\right), x \neq n\pi, n \in I$

64. ${}^{10}C_0\left[\text{coeff of } x^{10} \text{ in } (1+x)^{20}\right] - {}^{10}C_1\left[\text{coeff of } x^{10} \text{ in } (1+x)^{18}\right] + \ldots\ldots\ldots$

Coeff of x^{10} in $\left[{}^{10}C_0(1+x)^{20} - {}^{10}C_1(1+x)^{18} + \ldots\ldots\ldots\right]$

coeff of x^{10} in $\left[(1+x)^2 - 1\right]^{10}$

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coeff of x^{10} *in* $\left[x^{10}(2+x)^{10}\right]$

$= 2^{10}$

65. The given series is

$\cot^{-1}2+\cot^{-1}8+\cot^{-1}18+\cot^{-1}32+.....$

$=\cot^{-1}2(1)^2+\cot^{-1}2(2)^2+\cot^{-1}2(3)^2+\cot^{-1}2(4)^2+.....$

So, let $T_n=\cot^{-1}\left(2n^2\right)$

$=\tan^{-1}\dfrac{1}{2n^2}$

$=\tan^{-1}\left[\dfrac{(2n+1)-(2n-1)}{1+(2n+1)(2n-1)}\right]$

$=\tan^{-1}(2n+1)-\tan^{-1}(2n-1)$

$T_1=\tan^{-1}3-\tan^{-1}1$

$T_2=\tan^{-1}5-\tan^{-1}3$

$T_3=\tan^{-1}7-\tan^{-1}5\ldots$ and so on

$\therefore S_n=\tan^{-1}(2n+1)-\tan^{-1}1$

$\therefore \lim\limits_{n\to\infty}S_n=\tan^{-1}\infty-\dfrac{\pi}{4}$

$=\dfrac{\pi}{2}-\dfrac{\pi}{4}=\dfrac{\pi}{4}$

66. No. of elements in A = n^2 and of them n are of the form (x, x).

No. of good subsets = ${}^{n(n-1)}C_0+{}^{n(n-1)}C_1+.....+{}^{n(n-1)}C_{n(n-1)}=2^{n(n-1)}$

No. of reflexive relations = $2^{n(n-1)}$

67. $\sin\left(x+\dfrac{\pi}{12}\right)=\sin 2x$

68. $Let\ I=\int_{e^{-1}}^{e^2}\dfrac{|\log_e x|}{x}dx$

$=\int_{e^{-2}}^{1}\left|\dfrac{|\log_e x|}{x}\right|dx+\int_{1}^{e^2}\left|\dfrac{|\log_e x|}{x}\right|dx$

$=\int_{e^{-2}}^{1}\left(-\dfrac{|\log_e x|}{x}\right)dx+\int_{1}^{e^2}\dfrac{\log_e x}{x}dx$

Put $\log_e x=\Rightarrow\dfrac{dx}{x}=dz$

$\therefore I=\int_{-1}^{0}(-z)dz+\int_{0}^{2}z\,dz$

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$$=\left[\frac{-z^2}{2}\right]_{-1}^{0}+\left[\frac{z^2}{2}\right]_{0}^{2}$$

$$=-\frac{1}{2}[0-1]+\frac{1}{2}[4-0] \qquad =\frac{1}{2}+2=\frac{5}{2}$$

69. Conceptual

$$\lim_{x\to\infty}\frac{n\ln x-[x]}{[x]}=\lim_{x\to\infty}\frac{n\ln x-[x]}{[x]}-1=-1+\lim_{x\to\infty}\left[\frac{\frac{n\ln x}{x}}{\frac{[x]}{x}}\right]$$

70.

$$=-1+\lim_{x\to\infty}\left[\frac{\frac{n\ln x}{x}}{1-\frac{\{x\}}{x}}\right]=-1$$

Since $\lim_{x\to\infty}\frac{\ln x}{[x]}=\lim_{x\to\infty}\frac{1}{x}=0\Rightarrow\frac{\{x\}}{x}<\frac{1}{x}\to 0\,as\,x\to\infty$

71. k = 2,

Let $e^{i\frac{2\pi}{n}}=\alpha$ then $\sum_{j=1}^{n-12}\frac{1}{1-e^j}=\frac{1}{1-\alpha}+\frac{1}{1-\alpha^2}+.....+\frac{1}{1-\alpha^{n-1}}$

Where α is a nth root of unity $\left(\alpha,\alpha^2,\alpha^3,.....\alpha^{n-1}\right)$ are the roots of

$$\frac{x^n-1}{x-1}=(x-\alpha)\left(x-\alpha^2\right).....\left(x-\alpha^{n-1}\right)$$

Taking log on both side

$$\log\frac{x^n-1}{x-1}=\log(x-\alpha)+\log\left(x-\alpha^2\right)+.....+\log\left(x-\alpha^{n-1}\right)$$

Diff w.r.t. x and use $\lim_{x\to 1}$

$$\Rightarrow\frac{n-1}{2}=\frac{1}{1-\alpha}+\frac{1}{1-\alpha^2}+.....+\frac{1}{1-\alpha^{n-1}}$$

72. $y=m(x-3)\pm r\sqrt{1-m^2}$ & $t=mx+\frac{1}{m}$

$$\Rightarrow\pm r\sqrt{1-m^2}=3m+\frac{1}{m}\Rightarrow r^2=\frac{\left(3m^2+1\right)^2}{m^2\left(1-m^2\right)}$$

$\Rightarrow r^2\in[8,\alpha)$

$\Rightarrow\forall r<2\sqrt{2}$ No common tangents

$r=2\sqrt{2}\Rightarrow$ two common tangents

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$2\sqrt{2} < r < 3 \rightarrow$ four common tangents

$r = 3 \rightarrow$ three common tangents

$r > 3 \rightarrow$ two common tangents

73. It is the chord whose midpoint is (1, 2)

$\Rightarrow -x + y - 8 = 0 \quad [S_1 = S_{11}]$

74. $$\frac{C_K}{C_K - 1} = \frac{7 - K}{K}$$

$$\sum_{k=1}^{6} K\left\{7^2 - 14K + K^2\right\} = 49 \times 4 = 196$$

75. To get common terms in both the expansions

$$x^{r_1}\left(y^2\right)^{13-r_1} = \left(x^2\right)^{r_2}(y)^{14-r_2}$$

$r_1 = 2r_2$ & $26 - 2r_1 = 14 - r_2$

$r_1 = 8;\ r_2 = 4$

$\therefore$ Only one term is common

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